

- Gu, C., Zhang, Q., Grimstad, J., Hartman, A. J., Chadwick, M., Tomar, G. S., Garcia, X., Senter, E., Taropa, E., Pillai, T. S., Devlin, J., Laskin, M., de Las Casas, D., Valter, D., Tao, C., Blanco, L., Badia, A. P., Reitter, D., Chen, M., Brennan, J., Rivera, C., Brin, S., Iqbal, S., Surita, G., Labanowski, J., Rao, A., Winkler, S., Parisotto, E., Gu, Y., Olszewska, K., Zhang, Y., Addanki, R., Miech, A., Louis, A., Shafey, L. E., Teplyashin, D., Brown, G., Catt, E., Attaluri, N., Balaguer, J., Xiang, J., Wang, P., Ashwood, Z., Briukhov, A., Webson, A., Ganapathy, S., Sanghavi, S., Kannan, A., Chang, M.-W., Stjerngren, A., Djolonga, J., Sun, Y., Bapna, A., Aitchison, M., Pejman, P., Michalewski, H., Yu, T., Wang, C., Love, J., Ahn, J., Bloxwich, D., Han, K., Humphreys, P., Sellam, T., Bradbury, J., Godbole, V., Samangooei, S., Damoc, B., Kaskasoli, A., Arnold, S. M. R., Vasudevan, V., Agrawal, S., Riesa, J., Lepikhin, D., Tanburn, R., Srinivasan, S., Lim, H., Hodgkinson, S., Shyam, P., Ferret, J., Hand, S., Garg, A., Paine, T. L., Li, J., Li, Y., Giang, M., Neitz, A., Abbas, Z., York, S., Reid, M., Cole, E., Chowdhery, A., Das, D., Rogozińska, D., Nikolaev, V., Sprechmann, P., Nado, Z., Zilka, L., Prost, F., He, L., Monteiro, M., Mishra, G., Welty, C., Newlan, J., Jia, D., Allamanis, M., Hu, C. H., de Liedekerke, R., Gilmer, J., Saroufim, C., Rijhwani, S., Hou, S., Shrivastava, D., Baddepudi, A., Goldin, A., Ozturel, A., Cassirer, A., Xu, Y., Sohn, D., Sachan, D., Amplayo, R. K., Swanson, C., Petrova, D., Narayan, S., Guez, A., Brahma, S., Landon, J., Patel, M., Zhao, R., Villela, K., Wang, L., Jia, W., Rahtz, M., Giménez, M., Yeung, L., Lin, H., Keeling, J., Georgiev, P., Mincu, D., Wu, B., Haykal, S., Saputro, R., Vodrahalli, K., Qin, J., Cankara, Z., Sharma, A., Fernando, N., Hawkins, W., Neyshabur, B., Kim, S., Hutter, A., Agrawal, P., Castro-Ros, A., van den Driessche, G., Wang, T., Yang, F., Yiin Chang, S., Komarek, P., McIlroy, R., Lučić, M., Zhang, G., Farhan, W., Sharman, M., Natsev, P., Michel, P., Cheng, Y., Bansal, Y., Qiao, S., Cao, K., Shakeri, S., Butterfield, C., Chung, J., Rubenstein, P. K., Agrawal, S., Mensch, A., Soparkar, K., Lenc, K., Chung, T., Pope, A., Maggiore, L., Kay, J., Jhakra, P., Wang, S., Maynez, J., Phuong, M., Tobin, T., Tacchetti, A., Trebacz, M., Robinson, K., Katariya, Y., Riedel, S., Bailey, P., Xiao, K., Ghelani, N., Aroyo, L., Slone, A., Houlisby, N., Xiong, X., Yang, Z., Gribovskaya, E., Adler, J., Wirth, M., Lee, L., Li, M., Kagohara, T., Pavagadhi, J., Bridgers, S., Bortsova, A., Ghemawat, S., Ahmed, Z., Liu, T., Powell, R., Bolina, V., Iinuma, M., Zablotskaia, P., Besley, J., Chung, D.-W., Dozat, T., Comanescu, R., Si, X., Greer, J., Su, G., Polacek, M., Kaufman, R. L., Tokumine, S., Hu, H., Buchatskaya, E., Miao, Y., Elhawaty, M., Sidhant, A., Tomasev, N., Xing, J., Greer, C., Miller, H., Ashraf, S., Roy, A., Zhang, Z., Ma, A., Filos, A., Besta, M., Blevins, R., Klimenko, T., Yeh, C.-K., Changpinyo, S., Mu, J., Chang, O., Pajarskas, M., Muir, C., Cohen, V., Lan, C. L., Haridasan, K., Marathe, A., Hansen, S., Douglas, S., Samuel, R., Wang, M., Austin, S., Lan, C., Jiang, J., Chiu, J., Lorenzo, J. A., Sjöstrand, L. L., Cevey, S., Gleicher, Z., Avrahami, T., Boral, A., Srinivasan, H., Selo, V., May, R., Aisopos, K., Hussenot, L., Soares, L. B., Baumli, K., Chang, M. B., Recasens, A., Caine, B., Pritzel, A., Pavetic, F., Pardo, F., Gergely, A., Frye, J., Ramasesh, V., Horgan, D., Badola, K., Kassner, N., Roy, S., Dyer, E., Campos, V., Tomala, A., Tang, Y., Badawy, D. E., White, E., Mustafa, B., Lang, O., Jindal, A., Vikram, S., Gong, Z., Caelles, S., Hemsley, R., Thornton, G., Feng, F., Stokowiec, W., Zheng, C., Thacker, P., Çağlar Ünlü, Zhang, Z., Saleh, M., Svensson, J., Bileschi, M., Patil, P., Anand, A., Ring, R., Tshilas, K., Vezer, A., Selvi, M., Shevlane, T., Rodriguez, M., Kwiatkowski, T., Daruki, S., Rong, K., Dafoe, A., FitzGerald, N., Gu-Lemberg, K., Khan, M., Hendricks, L. A., Pellat, M., Feinberg, V., Cobon-Kerr, J., Sainath, T., Rauh, M., Hashemi, S. H., Ives, R., Hasson, Y., Li, Y., Noland, E., Cao, Y., Byrd, N., Hou, L., Wang, Q., Sottiaux, T., Paganini, M., Lespiau, J.-B., Moufarek, A., Hassan, S., Shivakumar, K., van Amersfoort, J., Mandhane, A., Joshi, P., Goyal, A., Tung, M., Brock, A., Sheahan, H., Misra, V., Li, C., Rakićević, N., Dehghani, M., Liu, F., Mittal, S., Oh, J., Noury, S., Sezener, E., Huot, F., Lamm, M., Cao, N. D., Chen, C., Elsayed, G., Chi, E., Mahdih, M., Tenney, I., Hua, N., Petrychenko, I., Kane, P., Scandinaro, D., Jain, R., Uesato, J., Datta, R., Sadovsky, A., Bunyan, O., Rabiej, D., Wu, S., Zhang, J., Vasudevan, G., Leurent, E., Alnahlawi, M., Georgescu, I., Wei, N., Zheng, I., Chan, B., Rabinovitch, P. G., Stanczyk, P., Zhang, Y., Steiner, D., Naskar, S., Azzam, M., Johnson, M., Paszke, A., Chiu, C.-C., Elias, J. S., Mohiuddin, A., Muhammad, F., Miao, J., Lee, A., Vieillard, N., Potluri, S., Park, J., Davoodi, E., Zhang, J., Stanway, J., Garmon, D., Karmarkar, A., Dong, Z., Lee, J., Kumar, A., Zhou, L., Evens, J., Isaac, W., Chen, Z., Jia, J., Levskaya, A., Zhu, Z., Gorgolewski, C., Grabowski, P., Mao, Y., Magni, A., Yao, K., Snider, J., Casagrande, N., Suganthan, P., Palmer, E., Irving, G., Loper, E., Faruqui, M., Arkatkar, I., Chen, N., Shafran, I., Fink, M., Castaño, A., Giannoumis, I., Kim, W., Rybiński, M., Sreevatsa, A., Prendki, J., Soergel, D., Goedeckemeyer, A., Gierke, W., Jafari, M., Gaba, M., Wiesner, J., Wright, D. G., Wei, Y., Vashisht, H., Kulizhskaya, Y., Hoover, J., Le, M., Li, L., Iwuanyanwu, C., Liu, L., Ramirez, K., Khorlin, A., Cui, A., LIN, T., Georgiev, M., Wu, M., Aguilar, R., Pallo, K., Chakladar, A., Repina, A., Wu, X., van der Weide, T., Ponnappalli, P., Kaplan, C., Simsa, J., Li, S., Dousse, O., Yang, F., Piper, J., Ie, N., Lui, M., Pasumarthi, R., Lintz, N., Vijayakumar, A., Thiet, L. N., Andor, D., Valenzuela, P., Paduraru, C., Peng, D., Lee, K., Zhang, S., Greene, S., Nguyen, D. D., Kurylowicz, P., Velury, S., Krause, S., Hardin, C., Dixon, L., Janzer, L., Choo, K., Feng, Z., Zhang, B., Singhal, A., Latkar, T., Zhang, M., Le, Q.,

- Abellan, E. A., Du, D., McKinnon, D., Antropova, N., Bolukbasi, T., Keller, O., Reid, D., Finchelstein, D., Raad, M. A., Crocker, R., Hawkins, P., Dadashi, R., Gaffney, C., Lall, S., Franko, K., Filonov, E., Bulanova, A., Leblond, R., Yadav, V., Chung, S., Askham, H., Cobo, L. C., Xu, K., Fischer, F., Xu, J., Sorokin, C., Alberti, C., Lin, C.-C., Evans, C., Zhou, H., Dimitriev, A., Forbes, H., Banarse, D., Tung, Z., Liu, J., Omernick, M., Bishop, C., Kumar, C., Sterneck, R., Foley, R., Jain, R., Mishra, S., Xia, J., Bos, T., Cideron, G., Amid, E., Piccinno, F., Wang, X., Banzal, P., Gurita, P., Noga, H., Shah, P., Mankowitz, D. J., Polozov, A., Kushman, N., Krakovna, V., Brown, S., Bateni, M., Duan, D., Firoiu, V., Thotakuri, M., Natan, T., Mohananey, A., Geist, M., Mudgal, S., Girgin, S., Li, H., Ye, J., Roval, O., Tojo, R., Kwong, M., Lee-Thorp, J., Yew, C., Yuan, Q., Bagri, S., Sinopalnikov, D., Ramos, S., Mellor, J., Sharma, A., Severyn, A., Lai, J., Wu, K., Cheng, H.-T., Miller, D., Sonnerat, N., Vnukov, D., Greig, R., Beattie, J., Caveness, E., Bai, L., Eisenschlos, J., Korchemniy, A., Tsai, T., Jasarevic, M., Kong, W., Dao, P., Zheng, Z., Liu, F., Yang, F., Zhu, R., Geller, M., Teh, T. H., Sanmiya, J., Gladchenko, E., Trdin, N., Sozanschi, A., Toyama, D., Rosen, E., Tavakkol, S., Xue, L., Elkind, C., Woodman, O., Carpenter, J., Papamakarios, G., Kemp, R., Kafle, S., Grunina, T., Sinha, R., Talbert, A., Goyal, A., Wu, D., Owusu-Afriyie, D., Du, C., Thornton, C., Pont-Tuset, J., Narayana, P., Li, J., Fatehi, S., Wieting, J., Ajmeri, O., Urias, B., Zhu, T., Ko, Y., Knight, L., Héliou, A., Niu, N., Gu, S., Pang, C., Tran, D., Li, Y., Levine, N., Stolovich, A., Kalb, N., Santamaria-Fernandez, R., Goenka, S., Yustalim, W., Strudel, R., Elqursh, A., Lakshminarayanan, B., Deck, C., Upadhyay, S., Lee, H., Dusenberry, M., Li, Z., Wang, X., Levin, K., Hoffmann, R., Holtmann-Rice, D., Bachem, O., Yue, S., Arora, S., Malmi, E., Mirylenka, D., Tan, Q., Koh, C., Yeganeh, S. H., Pöder, S., Zheng, S., Pongetti, F., Tariq, M., Sun, Y., Ionita, L., Seyedhosseini, M., Tafti, P., Kotikalapudi, R., Liu, Z., Gulati, A., Liu, J., Ye, X., Chrzaszcz, B., Wang, L., Sethi, N., Li, T., Brown, B., Singh, S., Fan, W., Parisi, A., Stanton, J., Kuang, C., Koverkathu, V., Choquette-Choo, C. A., Li, Y., Lu, T., Ittycheriah, A., Shroff, P., Sun, P., Varadarajan, M., Bahargam, S., Willoughby, R., Gaddy, D., Dasgupta, I., Desjardins, G., Cornero, M., Robenek, B., Mittal, B., Albrecht, B., Shenoy, A., Moiseev, F., Jacobsson, H., Ghafarkhah, A., Rivière, M., Walton, A., Crepy, C., Parrish, A., Liu, Y., Zhou, Z., Farabet, C., Radebaugh, C., Srinivasan, P., van der Salm, C., Fidljeland, A., Scellato, S., Latorre-Chimoto, E., Klimczak-Plucińska, H., Bridson, D., de Cesare, D., Hudson, T., Mendolicchio, P., Walker, L., Morris, A., Penchev, I., Mauger, M., Guseynov, A., Reid, A., Odoom, S., Loher, L., Cotruta, V., Yenugula, M., Grewe, D., Petrushkina, A., Duerig, T., Sanchez, A., Yadlowsky, S., Shen, A., Globerson, A., Kurzkrok, A., Webb, L., Dua, S., Li, D., Lahoti, P., Bhupatiraju, S., Hurt, D., Qureshi, H., Agarwal, A., Shani, T., Eyal, M., Khare, A., Belle, S. R., Wang, L., Tekur, C., Kale, M. S., Wei, J., Sang, R., Saeta, B., Liechty, T., Sun, Y., Zhao, Y., Lee, S., Nayak, P., Fritz, D., Vuyyuru, M. R., Aslanides, J., Vyas, N., Wicke, M., Ma, X., Bilal, T., Eltyshev, E., Balle, D., Martin, N., Cate, H., Manyika, J., Amiri, K., Kim, Y., Xiong, X., Kang, K., Luisier, F., Tripuraneni, N., Madras, D., Guo, M., Waters, A., Wang, O., Ainslie, J., Baldridge, J., Zhang, H., Pruthi, G., Bauer, J., Yang, F., Mansour, R., Gelman, J., Xu, Y., Polovets, G., Liu, J., Cai, H., Chen, W., Sheng, X., Xue, E., Ozair, S., Yu, A., Angermueller, C., Li, X., Wang, W., Wiesinger, J., Koukoumidis, E., Tian, Y., Iyer, A., Gurumurthy, M., Goldenson, M., Shah, P., Blake, M., Yu, H., Urbanowicz, A., Palomaki, J., Fernando, C., Brooks, K., Durden, K., Mehta, H., Momchev, N., Rahimtoroghi, E., Georgaki, M., Raul, A., Ruder, S., Redshaw, M., Lee, J., Jalan, K., Li, D., Perng, G., Hechtman, B., Schuh, P., Nasr, M., Chen, M., Milan, K., Mikulik, V., Strohman, T., Franco, J., Green, T., Hassabis, D., Kavukcuoglu, K., Dean, J., and Vinyals, O. Gemini: A family of highly capable multimodal models, 2023. URL <https://arxiv.org/abs/2312.11805>.
- Golkar, S., Pettee, M., Eickenberg, M., Bietti, A., Cranmer, M., Krawezik, G., Lanusse, F., McCabe, M., Ohana, R., Parker, L., Blancard, B. R.-S., Tesileanu, T., Cho, K., and Ho, S. xval: A continuous number encoding for large language models. *Neural Information Processing Systems (NeurIPS) AI for Science Workshop*, 2023. URL <https://openreview.net/forum?id=KHDMZtoF4i>.
- Goyal, S., Ji, Z., Rawat, A. S., Menon, A. K., Kumar, S., and Nagarajan, V. Think before you speak: Training language models with pause tokens. *International Conference on Learning Representations (ICLR)*, 2024. URL <https://openreview.net/forum?id=ph04CRkPdC>.
- Groeneveld, D., Beltagy, I., Walsh, P., Bhagia, A., Kinney, R., Tafjord, O., Jha, A. H., Ivison, H., Magnusson, I., Wang, Y., Arora, S., Atkinson, D., Authur, R., Chandu, K. R., Cohan, A., Dumas, J., Elazar, Y., Gu, Y., Hessel, J., Khot, T., Merrill, W., Morrison, J., Muennighoff, N., Naik, A., Nam, C., Peters, M. E., Pyatkin, V., Ravichander, A., Schwenk, D., Shah, S., Smith, W., Strubell, E., Subramani, N., Wortsman, M., Dasigi, P., Lambert, N., Richardson, K., Zettlemoyer, L., Dodge, J., Lo, K., Soldaini, L., Smith, N. A., and Hajishirzi, H. Olmo: Accelerating the science of language models, 2024.
- Hendrycks, D., Burns, C., Kadavath, S., Arora, A., Basart, S., Tang, E., Song, D., and Steinhardt, J. Measuring mathematical problem solving with the MATH dataset. *Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021. URL <https://openreview.net/forum?id=7Bywt2mQsCe>.

- Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., de las Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Lavaud, L. R., Lachaux, M.-A., Stock, P., Scao, T. L., Lavril, T., Wang, T., Lacroix, T., and Sayed, W. E. Mistral 7b. *arXiv:2310.06825*, 2023. URL <https://arxiv.org/abs/2310.06825>.
- Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., and Iwasawa, Y. Large language models are zero-shot reasoners. *Neural Information Processing Systems (NeurIPS)*, 2022. URL <https://openreview.net/forum?id=e2TBb5y0yFf>.
- Kudo, T. Subword regularization: Improving neural network translation models with multiple subword candidates. *Association for Computational Linguistics (ACL)*, 2018. URL <https://aclanthology.org/P18-1007>.
- Lanham, T., Chen, A., Radhakrishnan, A., Steiner, B., Denison, C., Hernandez, D., Li, D., Durmus, E., Hubinger, E., Kernion, J., et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv:2307.13702*, 2023. URL <https://arxiv.org/abs/2307.13702>.
- Lee, N., Sreenivasan, K., Lee, J. D., Lee, K., and Pappalopoulos, D. Teaching arithmetic to small transformers. *arXiv:2307.03381*, 2023. URL <https://arxiv.org/abs/2307.03381>.
- Lewkowycz, A., Andreassen, A. J., Dohan, D., Dyer, E., Michalewski, H., Ramasesh, V. V., Slone, A., Anil, C., Schlag, I., Gutman-Solo, T., Wu, Y., Neyshabur, B., Gur-Ari, G., and Misra, V. Solving quantitative reasoning problems with language models. *Neural Information Processing Systems (NeurIPS)*, 2022. URL <https://openreview.net/forum?id=IFXTZERXdm7>.
- Li, Y., Bubeck, S., Eldan, R., Del Giorno, A., Gunasekar, S., and Lee, Y. T. Textbooks Are All You Need II: phi-1.5 technical report. *arXiv preprint arXiv:2309.05463*, 2023.
- Liu, R., Garrette, D., Saharia, C., Chan, W., Roberts, A., Narang, S., Blok, I., Mical, R., Norouzi, M., and Constant, N. Character-aware models improve visual text rendering. *Association for Computational Linguistics (ACL)*, 2023. URL <https://aclanthology.org/2023.acl-long.900>.
- Lundberg, S. The art of prompt design: Prompt boundaries and token healing, 2023. URL <https://towardsdatascience.com/the-art-of-prompt-design-prompt-boundaries-and-token-healing-3b2448b0be38>.
- McCoy, R. T., Yao, S., Friedman, D., Hardy, M., and Griffiths, T. L. Embers of autoregression: Understanding large language models through the problem they are trained to solve. *arXiv:2309.13638*, 2023. URL <https://arxiv.org/abs/2309.13638>.
- Muffo, M., Cocco, A., and Bertino, E. Evaluating transformer language models on arithmetic operations using number decomposition. *Language Resources and Evaluation Conference (LREC)*, 2022. URL <https://aclanthology.org/2022.lrec-1.30>.
- Nogueira, R., Jiang, Z., and Lin, J. Investigating the limitations of transformers with simple arithmetic tasks. *arXiv:2102.13019*, 2021. URL <https://arxiv.org/abs/2102.13019>.
- Nye, M., Andreassen, A. J., Gur-Ari, G., Michalewski, H., Austin, J., Bieber, D., Dohan, D., Lewkowycz, A., Bosma, M., Luan, D., et al. Show your work: Scratchpads for intermediate computation with language models. *arXiv:2112.00114*, 2021. URL <https://arxiv.org/abs/2112.00114>.
- OpenAI, Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., Avila, R., Babuschkin, I., Balaji, S., Balcom, V., Baltescu, P., Bao, H., Bavarian, M., Belgum, J., Bello, I., Berdine, J., Bernadett-Shapiro, G., Berner, C., Bogdonoff, L., Boiko, O., Boyd, M., Brakman, A.-L., Brockman, G., Brooks, T., Brundage, M., Button, K., Cai, T., Campbell, R., Cann, A., Carey, B., Carlson, C., Carmichael, R., Chan, B., Chang, C., Chantzis, F., Chen, D., Chen, S., Chen, R., Chen, J., Chen, M., Chess, B., Cho, C., Chu, C., Chung, H. W., Cummings, D., Currier, J., Dai, Y., Decareaux, C., Degry, T., Deutsch, N., Deville, D., Dhar, A., Dohan, D., Dowling, S., Dunning, S., Ecoffet, A., Eleti, A., Eloundou, T., Farhi, D., Fedus, L., Felix, N., Fishman, S. P., Forte, J., Fulford, I., Gao, L., Georges, E., Gibson, C., Goel, V., Gogineni, T., Goh, G., Gontijo-Lopes, R., Gordon, J., Grafstein, M., Gray, S., Greene, R., Gross, J., Gu, S. S., Guo, Y., Hallacy, C., Han, J., Harris, J., He, Y., Heaton, M., Heidecke, J., Hesse, C., Hickey, A., Hickey, W., Hoeschele, P., Houghton, B., Hsu, K., Hu, S., Hu, X., Huizinga, J., Jain, S., Jain, S., Jang, J., Jiang, A., Jiang, R., Jin, H., Jin, D., Jomoto, S., Jonn, B., Jun, H., Kaftan, T., Łukasz Kaiser, Kamali, A., Kanitscheider, I., Keskar, N. S., Khan, T., Kilpatrick, L., Kim, J. W., Kim, C., Kim, Y., Kirchner, H., Kiros, J., Knight, M., Kokotajlo, D., Łukasz Kondraciuk, Kondrich, A., Konstantinidis, A., Kopic, K., Krueger, G., Kuo, V., Lampe, M., Lan, I., Lee, T., Leike, J., Leung, J., Levy, D., Li, C. M., Lim, R., Lin, M., Lin, S., Litwin, M., Lopez, T., Lowe, R., Lue, P., Makanju, A., Malfacini, K., Manning, S., Markov, T., Markovski, Y., Martin, B., Mayer, K., Mayne, A., McGrew, B., McKinney, S. M., McLeavey, C., McMillan, P., McNeil, J., Medina, D., Mehta, A., Menick, J., Metz, L., Mishchenko, A., Mishkin, P., Monaco, V., Morikawa, E., Mossing, D., Mu, T., Murati, M., Murk, O.,